Are You Getting What You Pay For? Auditing Model Substitution in LLM APIs

Why software-only audits fail under realistic conditions — and why hardware attestation wins

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Audience: ML researchers • AI security practitioners • Cloud infrastructure engineers

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Core trust gap: the API is a black box — users cannot verify the advertised model is served.

Providers face strong incentives to reduce GPU cost while keeping pricing constant.

Common substitution types

1) Smaller model
e.g., serve 8B instead of 70B

2) Quantized variant
e.g., FP8/INT8 instead of FP16

3) Fine-tuned / updated
behavior shifts without disclosure

4) Different family
swap architecture / provider

Formal framing: hypothesis testing

H0 (specified model)
Samples follow P_spec(y | x)

H1 (substituted model)
Samples follow P_actual(y | x)

Goal: determine whether an auditor can distinguish P_spec from P_actual given only API samples.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Benchmark deltas from FP16 → FP8 are often within error bars — so output-only audits struggle to separate specified vs. substituted deployments.

Model family	MMLU Orig	MMLU FP8	GSM8K Orig	GSM8K FP8	MATH Orig	MATH FP8	GPQA D Orig	GPQA D FP8
Llama-3-8B	62.69 ± 0.18	62.43 ± 0.26	61.14 ± 3.47	60.90 ± 4.10	20.65 ± 3.43	14.91 ± 2.49	22.62 ± 0.22	20.14 ± 0.24
Llama-3-70B	78.05 ± 0.08	77.88 ± 0.13	88.06 ± 1.44	87.35 ± 1.24	35.69 ± 1.33	35.75 ± 1.16	29.60 ± 0.51	33.12 ± 0.30
Gemma-2-9B	71.86 ± 0.08	71.92 ± 0.11	81.80 ± 1.35	79.41 ± 1.14	33.41 ± 0.28	32.53 ± 0.34	28.64 ± 2.97	27.81 ± 3.22
Qwen2-72B	82.18 ± 0.08	81.98 ± 0.08	86.72 ± 1.00	86.82 ± 0.97	37.39 ± 1.41	37.67 ± 1.39	29.93 ± 2.71	31.08 ± 1.96
Mistral-7B	59.15 ± 0.10	58.77 ± 0.13	35.90 ± 4.54	32.20 ± 4.02	8.94 ± 1.24	7.68 ± 1.23	21.60 ± 0.17	22.72 ± 0.19

Takeaway: small score shifts enable cost-saving substitutions with minimal external signal.

Higher Temperature Amplifies Performance Gaps but Increases Noise

Temperature scaling experiments (T = 0.2, 0.5, 1.0, 2.0) show that raising temperature degrades accuracy for all models, while the substitution signal becomes noisier.

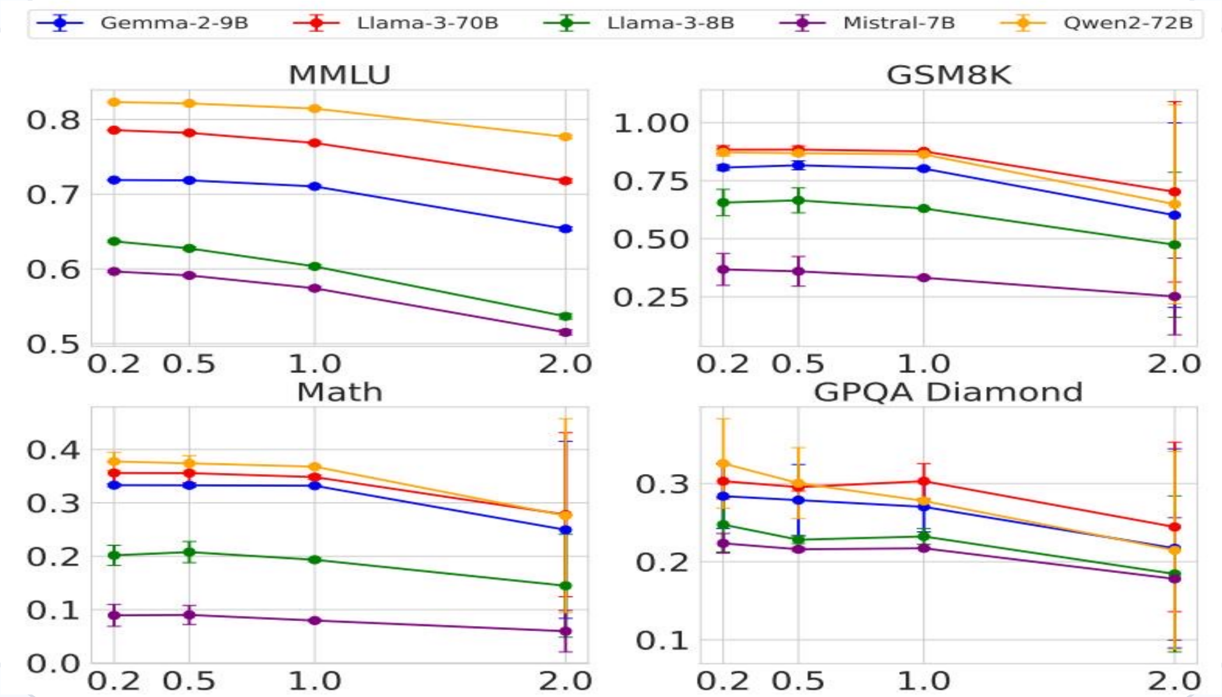


Figure 1: Benchmark accuracy vs. temperature (four tasks × five models)

Interpretation

- At high temperature (e.g., T = 2.0) all models degrade.
- As outputs diversify, audit statistics become higher-variance.
- Gaps between original and substitute can narrow or become

Audit implication

Increasing temperature can accentuate behavioral differences, but also increases randomness — shrinking practical power of statistical tests.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Token-level log-prob fingerprints break in production: the same model can emit different log probabilities across inference stacks and hardware.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: Log-prob variation for Llama-3-8B across software versions and GPUs

Observed nondeterminism

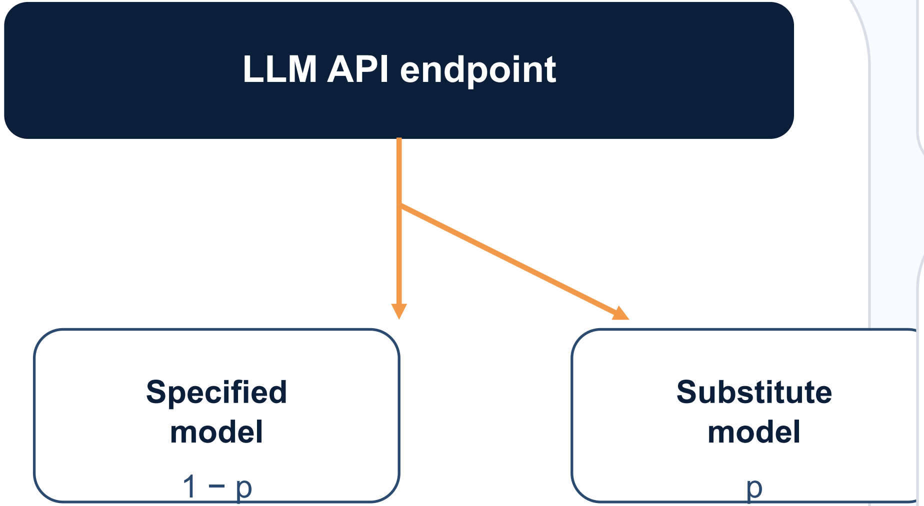
- Framework: transformers vs vLLM
- Version drift: transformers 4.40 vs 4.50; vLLM 0.6.2 vs 0.8.2
- Hardware: A100 vs H100

Audit consequence

Differences in log probabilities can reflect benign deployment changes, not substitution — defeating metadata-based detection in practice.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Attack: randomized substitution



As $p \to 0$, the mixed output distribution converges to the specified model's distribution — making detection increasingly query-intensive.

Why detection breaks

P_mix = (1 − p)·P_spec + p·P_sub
As p decreases, P_mix becomes harder to distinguish from P_spec.

Software-only tests become query-prohibitive:
• Low-rate routing hides the substitute signal• Large query budgets are required to reach power• Providers can combine with benchmark evasion

Software-Only Auditing Methods Are Fundamentally Unreliable

Comparison matrix of key failure modes described in the source brief.

Method type	Primary weakness	Practical limitation
Text-based statistical tests	Require impractically large query budgets; subtle substitutions can fall within noise.	Query-prohibitive; power collapses under randomized substitution.
Log-probability fingerprinting	Production inference is nondeterministic across frameworks/versions/hardware.	Cannot separate benign deployment drift from substitution.
Adversarial countermeasures	Provider can actively evade audits (randomized routing, benchmark evasion).	Audits become gameable; results are not actionable proofs.

Bottom line: software-only evidence is not a verifiable proof of served model identity.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEE-based verification (proposed)

TEE = hardware-secured enclave + cryptographic attestation.

Provides proof that:
- Specific inference code is running
- Specific model weights are loaded
- Execution occurs inside the enclave

Enables verifiable inference for black-box APIs.

Why TEEs win vs software-only

Provable
Cryptographic attestation (not statistical evidence).

Efficient
Modest performance overhead (per brief).

Robust
Unaffected by temperature, nondeterminism, or routing tricks.

Deployable
Practical path: NVIDIA confidential computing (H100).

Positioning: unlike ZK proofs (computationally prohibitive for large models), TEEs are practical and deployable today.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1. Model substitution is economically motivated and hard to audit from black-box outputs.

2. FP8 quantization barely shifts benchmark scores, enabling subtle cost-saving swaps.

3. Inference nondeterminism defeats log-probability fingerprinting in production.

4. Randomized routing + benchmark evasion makes statistical detection query-prohibitive.

5. Trusted Execution Environments (e.g., NVIDIA confidential computing) provide actionable, provable integrity.

Actionable path

Shift from
statistical suspicion
→ cryptographic proof

Serve models inside
TEEs and
attest weights + code.

From the source brief

TEEs provide
provable, efficient,
and robust model
integrity guarantees
— the most
actionable solution
available today.